

PHYS 1B- Summer Session I (2024)

Lesson plan for lectures, quizzes and HW (tentative, will be edited as the course progresses)

Last updated: 2.19 pm, Aug 1, 2024

The text in **red** is tentative and the text in black is in its final version.

(We might not be able to cover all the topics mentioned in the plan for the lectures. This document will be updated after every lecture.)

Lectures:

Lecture number	Date	Topics covered in lecture and lecture notes	Relevant sections in textbook (Kudu)
1	07/01/2024 (Week 1)	Introduction to the course, Electric charges- a few definitions, properties of electric charge, qualitative properties of electric force, charge carriers (electrons and protons), conductors and insulators, charging by friction, charging by contact and induced polarization	Section B01- subsections B01.1, B01.2, B01.3, B01.4, B01.5, B01.6
2	07/02/2024 (Week 1)	Charging by induction, Coulomb's law, Principle of superposition, solving problems.	Section B01- subsections B01.7, B01.8, B01.9.
3	07/03/2024 (Week 1)	Principle of superposition (problems), Electric field, electric field of a point charge, superposition of fields, electric force exerted by a field, electric field for a 1d arrangement of charges, solving problems.	Section B01- subsections B01.9. Section B02- subsections B02.1, B02.2
4	07/08/2024 (Week 2)	Electric field for a 2d arrangement of charges, solving problems. Electric field lines (introduction).	Section B02- subsections B02.2, B02.3.
5	07/09/2024	Electric field lines, Field lines for	Section B02-

	(Week 2)	multiple charges, field lines corresponding to symmetric charge distributions, field due to infinite planes (introduction to parallel plate capacitors). Motion of a charged particle in an E-field, solving problems.	subsections B02.3, B02.4, B02.5 (field lines for symmetric charge distributions), B02.6, B02.7
6	07/11/2024 (Week 2)	Area vector, electric flux. Gauss' Law, Applying Gauss' law, Charge Distribution with Cylindrical Symmetry (infinite line of uniform charge).	Section B03- subsections B03.1-B03.8, B03.10.
7	07/15/2024 (Week 3)	Review of material from week 2. Charge Distribution with Spherical Symmetry, Charge distribution with Planar symmetry. Conductors and Electric Fields, Conductors and excess charge (conductors vs insulators), electric field inside a conductor.	Relevant sections from Week 2 (Lectures 4, 5 and 6). Section B03- subsections B03.9, B03.10, B03.11, B03.12, B03.13.
8	07/16/2024 (Week 3)	Review of electric field due to an infinite line of uniform charge, infinite plane of uniform charge, charge distributions with spherical symmetry. Review of conductors. Solving problems. Electric potential energy. Electric potential, Electric potential due to a point charge, Potential Difference.	Section B03- B03.9, B03.12, B03.13 Section B04- subsections B04.1, B04.2, B04.3, B04.7.
9	07/18/2024 (Week 3)	Summary of previous lectures. Electron volt, Electric potential in	Section B04- subsections B04.4,

		a uniform E-field, potential energy and kinetic energy, Potential due to multiple charges, Equipotential lines and surfaces. Parallel plates (intro to capacitors). Capacitors and Capacitance, calculation of capacitance, parallel plate capacitor.	B04.5, B04.6, B04.7, B04.8 Section B05- subsections B05.1, B05.2, B05.3, B05.4.
10	07/22/2024 (Week 4)	Review. Energy stored in a capacitor. Capacitors in series and parallel, series combination, parallel combination.	Relevant sections from week 3. Section B05- subsections B05.6, B05.7, B05.8, B05.9.
11	07/23/2024 (Week 4)	Review. Capacitor with a dielectric. Solving problems. Electric current, emf, drift velocity, Ohm's law. Making electrical measurements (voltage-voltmeter and current-ammeter, see B07.18 and B07.23). Electrical resistance. Resistivity and resistance.	Section B05- B05.10, B05.11. Section B06- Subsections B06.1, B06.2, B06.3, B06.4, B06.5 Section B07- B07.8. Section B07- B07.18, B07.23. Section B06-B06.6.
12	07/25/2024 (Week 4)	Review. Solving problems on capacitors. Resistors in series, resistors in parallel, combinations of series and parallel, equivalent resistance. Electric power. Solving problems on resistors.	B06.8 Section B07- Subsections B07.1, B07.2, B07.3, B07.4, B07.5, B07.7, B07.8, B07.11, B07.12, B07.13, B07.14, B07.15, B07.16

		Kirchoff's rules (procedure summary), Kirchoff's first rule, Kirchoff's second rule, applying Kirchoff's rule, practice on a circuit.	
13	07/29/2024 (Week 5)	Review, understanding Kirchoff's rules, power rating, sources of potential difference, circuit symbols. RC circuits (time dependent circuits), charging a capacitor, discharging a capacitor. Introduction to magnetism, magnets.	Relevant sections from Week 4. B07.20, B07.21, B07.22. Section B08- subsections B08.1, B08.2, B08.5, B08.6.
14	07/30/2024 (Week 5)	Review- RC circuits (time dependent circuits), charging a capacitor, discharging a capacitor. Ferromagnets, magnetism. Review of vector product. Magnetic field, magnetic field lines. Magnetic field due to a straight current carrying conductor. Electric currents and magnetism. Lorentz force.	B07.20-B07.22 B08.1-B08.7, B08.11
15	07/31/2024 (Week 5)	Solving problems- RC circuits, right hand rule, magnetic force. Mass spectrometer. Magnetic field due to a current carrying loop. Magnetic flux, Lenz's law. Solving problems.	B07.20-B07.22 B08.7, B08.9, B08.12, B08.11. B09.1, B09.4, B09.6, B09.9, B09.10. B10.1, B10.2, B10.4, B10.5.

16	08/01/2024 (Week 5)	Review. Introduction to- Forces on wires, motional EMF, Generator, transformers, magnetic moment, torque on a loop, motor, inductors, Ampere's law, Maxwell's equations, electromagnetic waves, superconductors, AC circuits, LCR circuits.	
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Quizzes:

Quiz number	Date	Topics covered in lectures and lecture notes	Sections in textbook (Kudu)
1	07/10/2024 (Week 2)	Topics from lectures 1, 2, 3 Topics from lecture 4 (up until the point mentioned in the lecture)	Section B01- subsections B01.1-B01.9 Section B02- subsections B02.1, B02.2
2	07/17/2024 (Week 3)	Topics from lectures 4, 5, 6. Topics from lecture 7 (up until the point mentioned in the lecture)	Section B02- subsections B02.2, B02.3. Section B02- subsections B02.3, B02.4, B02.5 (field lines for symmetric charge distributions), B02.6, B02.7
3	07/24/2024 (Week 4)	Topics from lectures 7, 8, 9 Topics from lecture 10 (up until the point mentioned in the lecture)	Relevant sections from Week 2 (Lectures 4, 5 and 6). Section B03- subsections B03.9, B03.10, B03.11, B03.12,

			B03.13. Section B03- B03.9, B03.12, B03.13 Section B04- subsections B04.1, B04.2, B04.3, B04.7. Section B04- subsections B04.4, B04.5, B04.6, B04.7, B04.8 Section B05- subsections B05.1, B05.2, B05.3, B05.4.
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Kudu HW:

Kudu HW number	Submission deadline	Topics covered in lectures and lecture notes	Sections in textbook (Kudu)
1	07/08/2024 (Week 2)	Topics from lectures 1, 2, 3	Section B01- subsections B01.1-B01.9 Section B02- subsections B02.1, B02.2
2	07/15/2024 (Week 3)	Topics from lectures 4, 5, 6	Section B02- subsections B02.2, B02.3, B02.4, B02.5 (field lines for symmetric charge distributions), B02.6, B02.7 Section B03- subsections B03.1-B03.8, B03.10.
3	07/23/2024 (Week 4)	Topics from lectures 7, 8, 9	Relevant sections from Week 2 (Lectures 4, 5

			and 6). Section B03- subsections B03.9, B03.10, B03.11, B03.12, B03.13. Section B04- subsections B04.1, B04.2, B04.3, B04.4, B04.5, B04.6, B04.7, B04.8. Section B05- subsections B05.1, B05.2, B05.3, B05.4
4	07/30/2024 (Week 5)	Topics from lectures 10, 11, 12, 13	Relevant sections from week 3. Section B05- subsections B05.6, B05.7, B05.8, B05.9. Section B05- B05.10, B05.11. Section B06- Subsections B06.1, B06.2, B06.3, B06.4, B06.5 Section B07- B07.8. Section B07- B07.18, B07.23. Section B06-B06.6. B06.8 Section B07- Subsections B07.1, B07.2, B07.3, B07.4, B07.5, B07.7, B07.8, B07.11, B07.12, B07.13,

			B07.14, B07.15, B07.16 B07.20, B07.21, B07.22
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Descriptive HW-draft version: (see calendar for submission deadline for the Descriptive HW- corrected version)

Descriptive HW number	Submission deadline for draft version	Topics covered in lectures and lecture notes	Sections in textbook (Kudu)
1	07/04/2024 (Week 1)	Topics from lectures 1, 2, 3	Section B01- subsections B01.1-B01.9 Section B02- subsections B02.1, B02.2
2	07/12/2024 (Week 2)	Topics from lectures 3, 4, 5	Section B01- subsections B01.9. Section B02- subsections B02.1, B02.2, B02.3, B02.4, B02.5 (field lines for symmetric charge distributions), B02.6, B02.7. Section B03- subsections B03.1-B03.8, B03.10.
3	07/19/2024 (Week 3)	Topics from lectures 6, 7, 8, 9	Relevant sections from Week 2 (Lectures 4, 5 and 6).

			Section B03- subsections B03.1-B03.8, B03.10. Section B03- subsections B03.9, B03.10, B03.11, B03.12, B03.13. Section B04- subsections B04.1, B04.2, B04.3, B04.4, B04.5, B04.6, B04.7, B04.8. Section B05- subsections B05.1, B05.2, B05.3, B05.4
4	07/30/2024 (Week 4)	Topics from lectures 9, 10, 11, 12	Section B04- subsections B04.4, B04.5, B04.6, B04.7, B04.8 Section B05- subsections B05.1, B05.2, B05.3, B05.4. Relevant sections from week 3. Section B05- subsections B05.6, B05.7, B05.8, B05.9. Section B05- B05.10, B05.11. Section B06- Subsections B06.1, B06.2, B06.3, B06.4, B06.5 Section B07- B07.8.

			Section B07- B07.18, B07.23. Section B06-B06.6, B06.8 Section B07- Subsections B07.1, B07.2, B07.3, B07.4, B07.5, B07.7, B07.8, B07.11, B07.12, B07.13, B07.14, B07.15, B07.16
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Canvas practice questions:

During week 5, a set of practice questions will be posted on Canvas which will be graded based on participation which will contain problems based on Lectures 13, 14, and 15 so that students can review the contents of lectures that happened during week 5. (Submission deadline: 11.59 pm, 07/31/2024)